SMARTX: Porewater Ammonium

May 2025 Samples

2026-08-31

Contents

##Run Information

```r
cat("Run Information:  ") #lets you know what section you're in
```

```
## Run Information:
```

```r
#set the run date & user name
  run_date <- "08/12/2025"
  sample_year <- "2025"
  sample_month <- "May"
  user <- "Victoria Ellis"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/SMARTX_May_2025_Run1.csv",
                 "Raw Data/SMARTX_May_2025_Run2.csv",
                 "Raw Data/SMARTX_May_2025_Run3.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- "Raw Data/SEAL_SMARTX_QAQC_Log_2025.csv"
  final_path <- "Processed Data/GCReW_SMARTX_Porewater_NH4_202505.csv"

#record any notes about the run or anything other info here:
  run_notes <- "PE Check contaminated, NH3 high, PE Check changed after this run
               Moved metadata chunk to setup because 'here' package was needed"

  #Add SMARTX Prefix if necessary to samples
  Add_Prefix <- "YES"


  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

```
## PE Check contaminated, NH3 high, PE Check changed after this run
##                  Moved metadata chunk to setup because 'here' package was needed
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

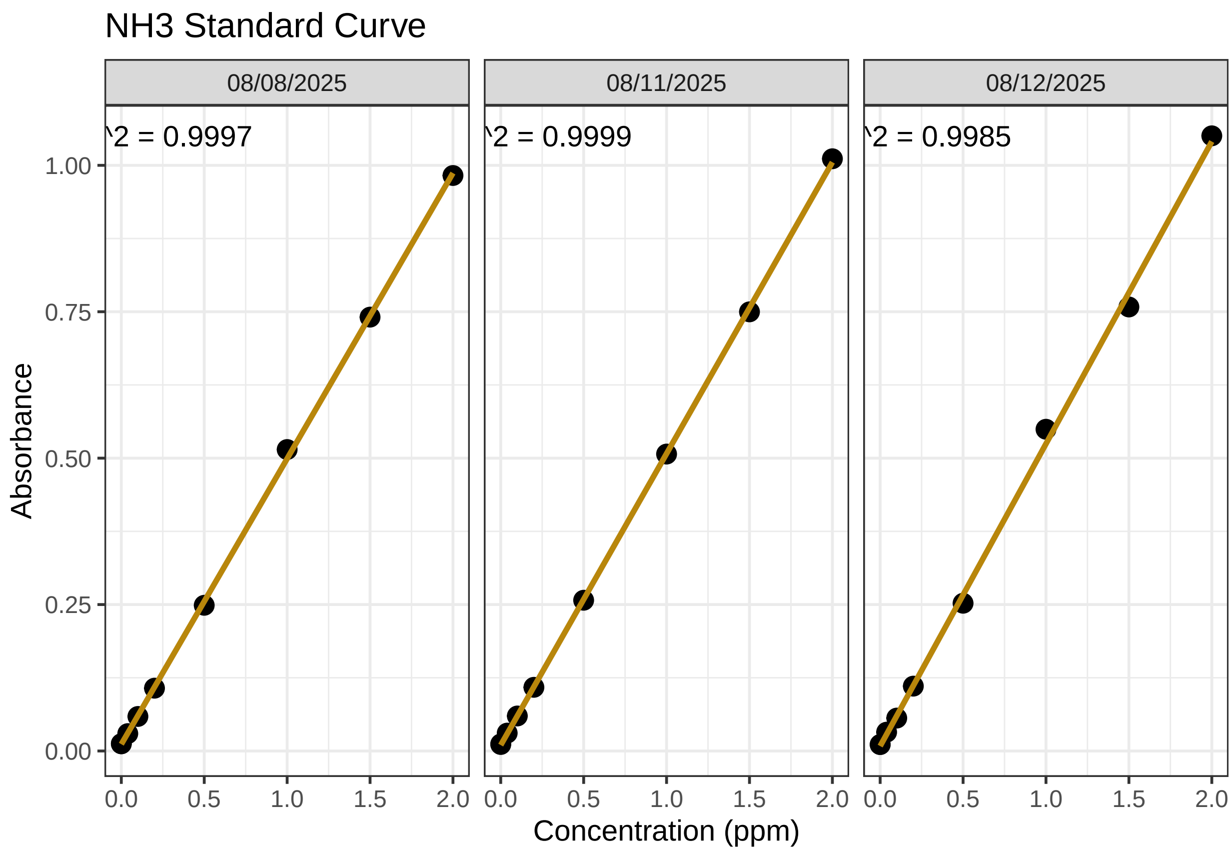
0.1 Import Data & Clean

0.2 Assessing standard Curves

#Pull out standards data

```
## Assess Standard Curves
```

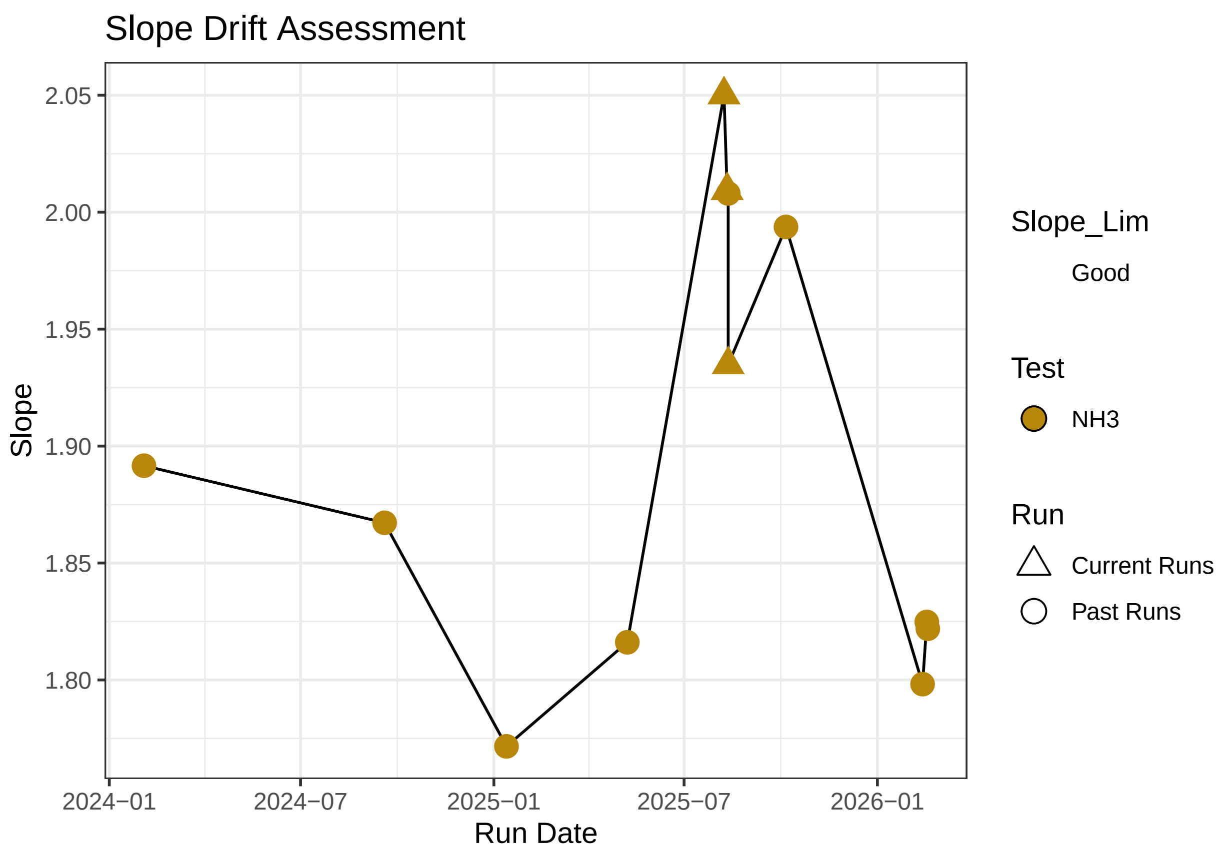
```
## `geom_smooth()` using formula = 'y ~ x'
```



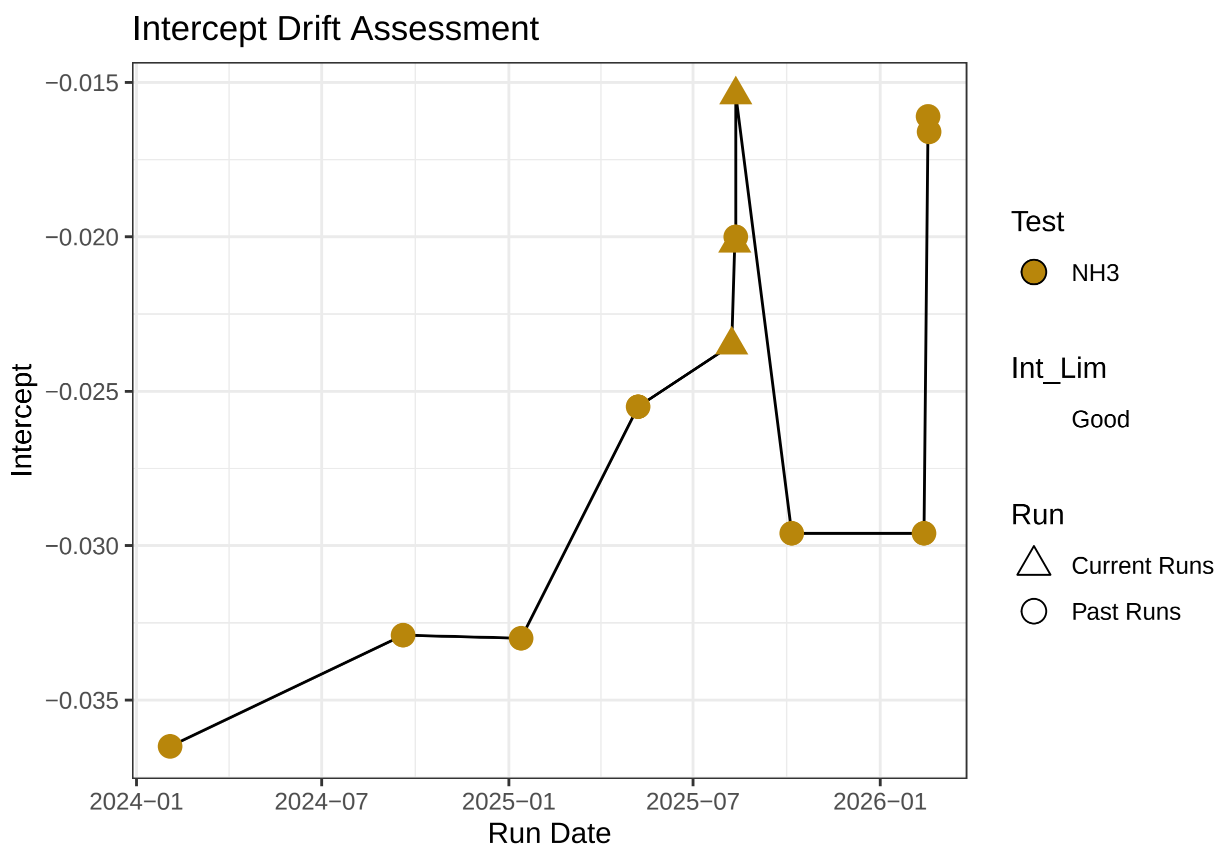
Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	08/08/2025	R2_Pass	0.9996549	2.050170
Ammonia 2	08/11/2025	R2_Pass	0.9999057	2.009256
Ammonia 2	08/12/2025	R2_Pass	0.9984853	1.934847

#Plot standards data

```
## `summarise()` has grouped output by 'Test', 'Intercept', 'Slope', 'R_Squared',
## 'Run_Date'. You can override using the `.groups` argument.
```



Slope Drift Assessment
2.05
2.00
1.95
1.90
1.85
1.80
Slope
2024−01
2024−07
2025−01
2025−07
2026−01
Run Date
Slope_Lim
Good
Test
NH3
Run
Current Runs
Past Runs



Intercept Drift Assessment
−0.015
−0.020
−0.025
−0.030
−0.035
Intercept
2024−01
2024−07
2025−01
2025−07
2026−01
Run Date
Test
NH3
Int_Lim
Good
Run
Current Runs
Past Runs

0.3 Dilution Corrections - ensure the latest dilution is kept

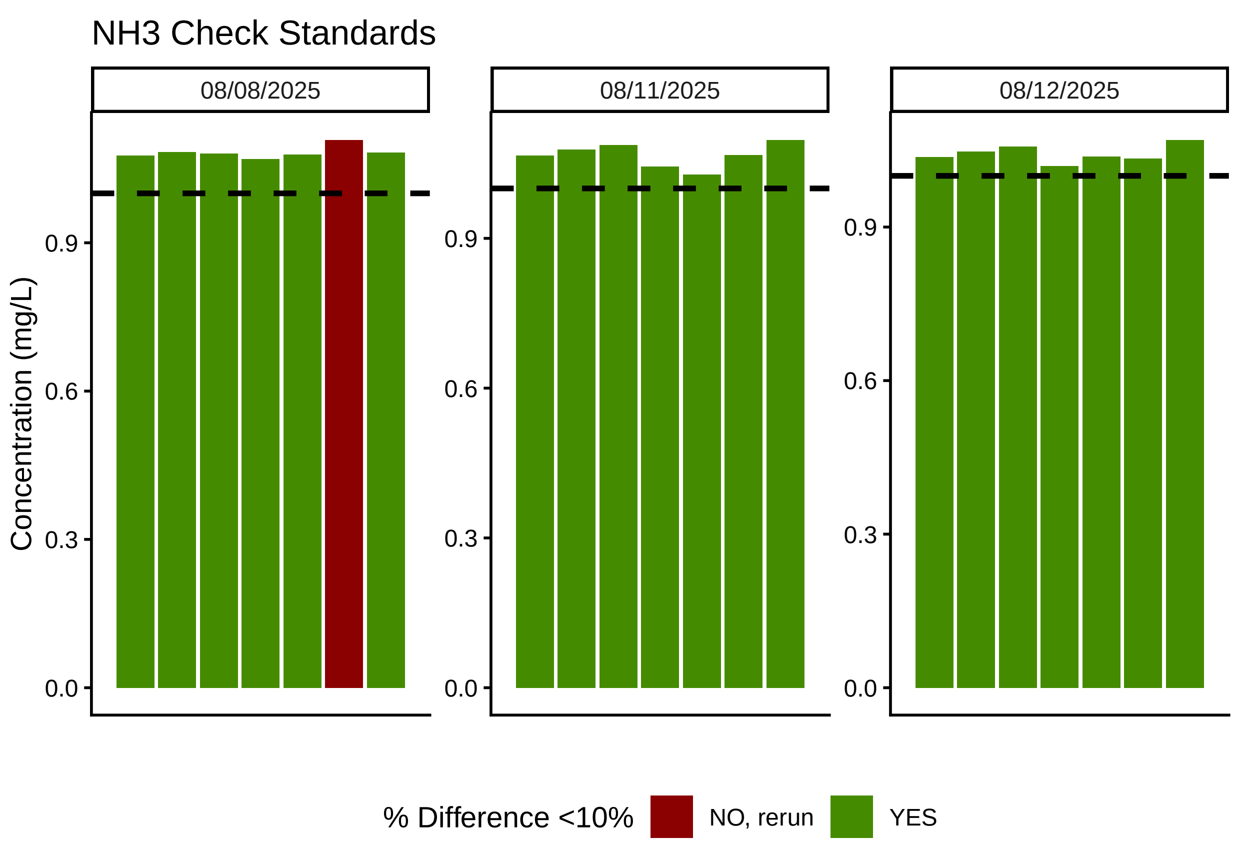
```
## [1] "Reran Samples: c(\"442-40\", \"442-80\", \"443-40\", \"443-80\", \"443-120\", \"450-40\", \"450-

## [1] "Diluted Samples: c(\"442-40\", \"442-80\", \"443-40\", \"443-80\", \"443-120\", \"450-40\", \"45

## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	08/08/2025	Performance Check NOT Within 25% - REASSESS	1.500342	1.034
Ammonia 2	08/11/2025	Performance Check Within 25% - PROCEED	1.277440	1.034
Ammonia 2	08/12/2025	Performance Check NOT Within 25% - REASSESS	1.443127	1.034

0.5 Analyze the Check Standards

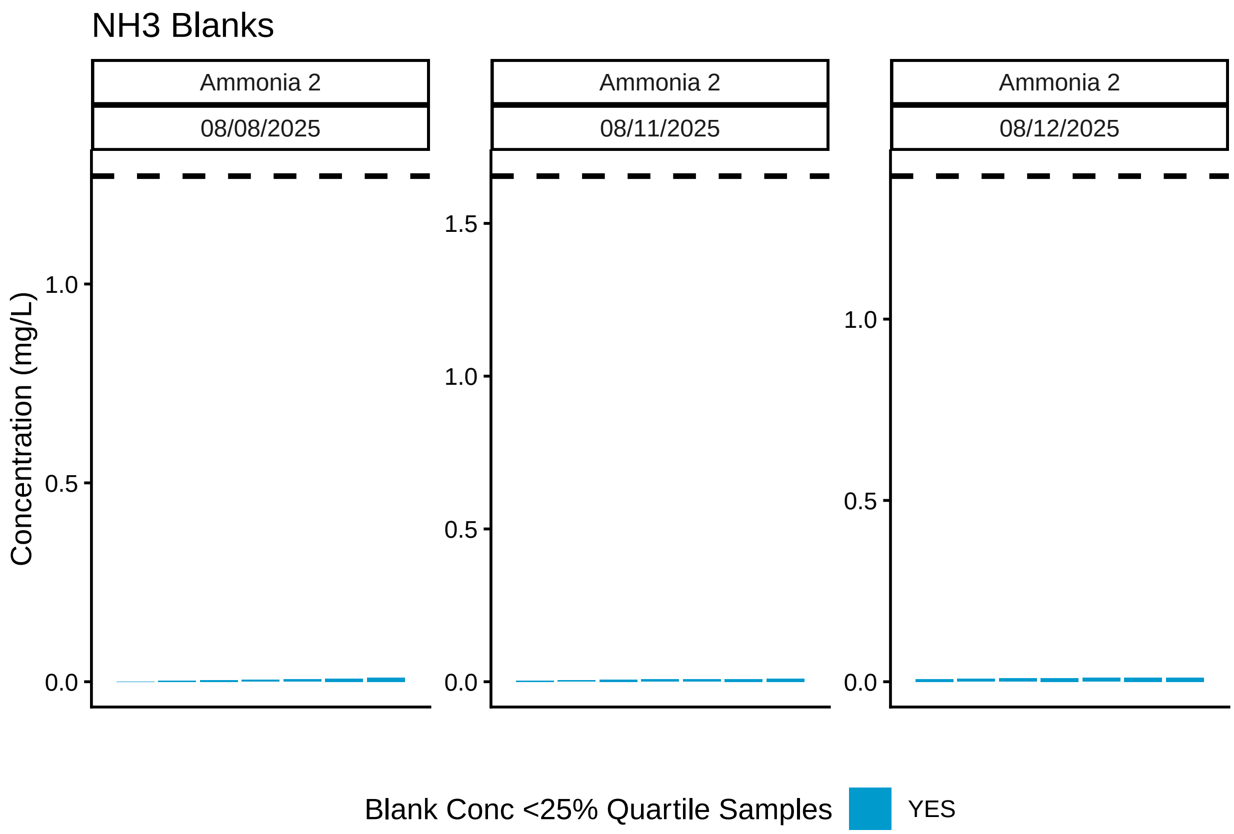
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	08/08/2025	RSV WITHIN RANGE - PROCEED	0.0225172	0.25
Ammonia 2	08/11/2025	RSV WITHIN RANGE - PROCEED	0.0225172	0.25
Ammonia 2	08/12/2025	RSV WITHIN RANGE - PROCEED	0.0225172	0.25



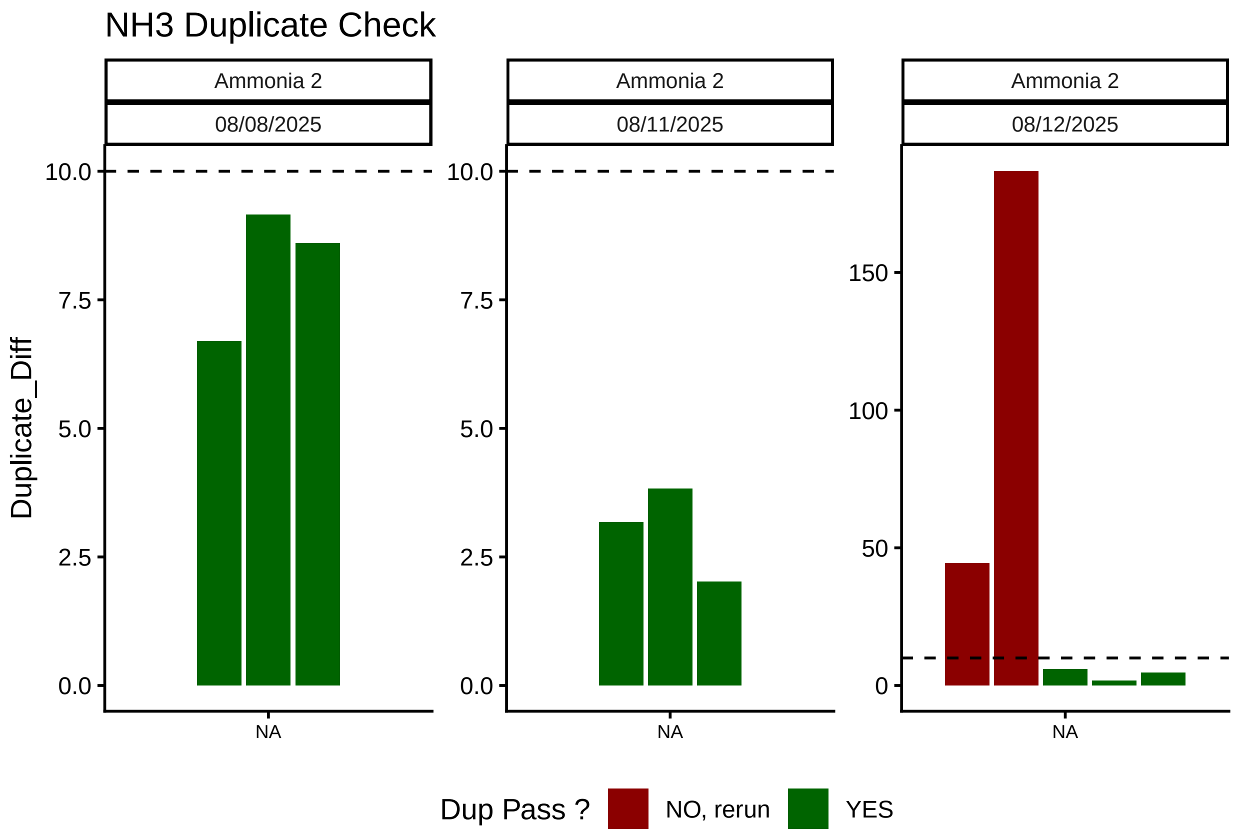
Test	Run_Date	CHK_Flag
Ammonia 2	08/08/2025	>60% of Checks Pass - PROCEED
Ammonia 2	08/11/2025	>60% of Checks Pass - PROCEED
Ammonia 2	08/12/2025	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	08/08/2025	>60% of Blanks Pass - PROCEED	0.0052279	1.271356
Ammonia 2	08/11/2025	>60% of Blanks Pass - PROCEED	0.0072246	1.654718
Ammonia 2	08/12/2025	>60% of Blanks Pass - PROCEED	0.0098191	1.394393

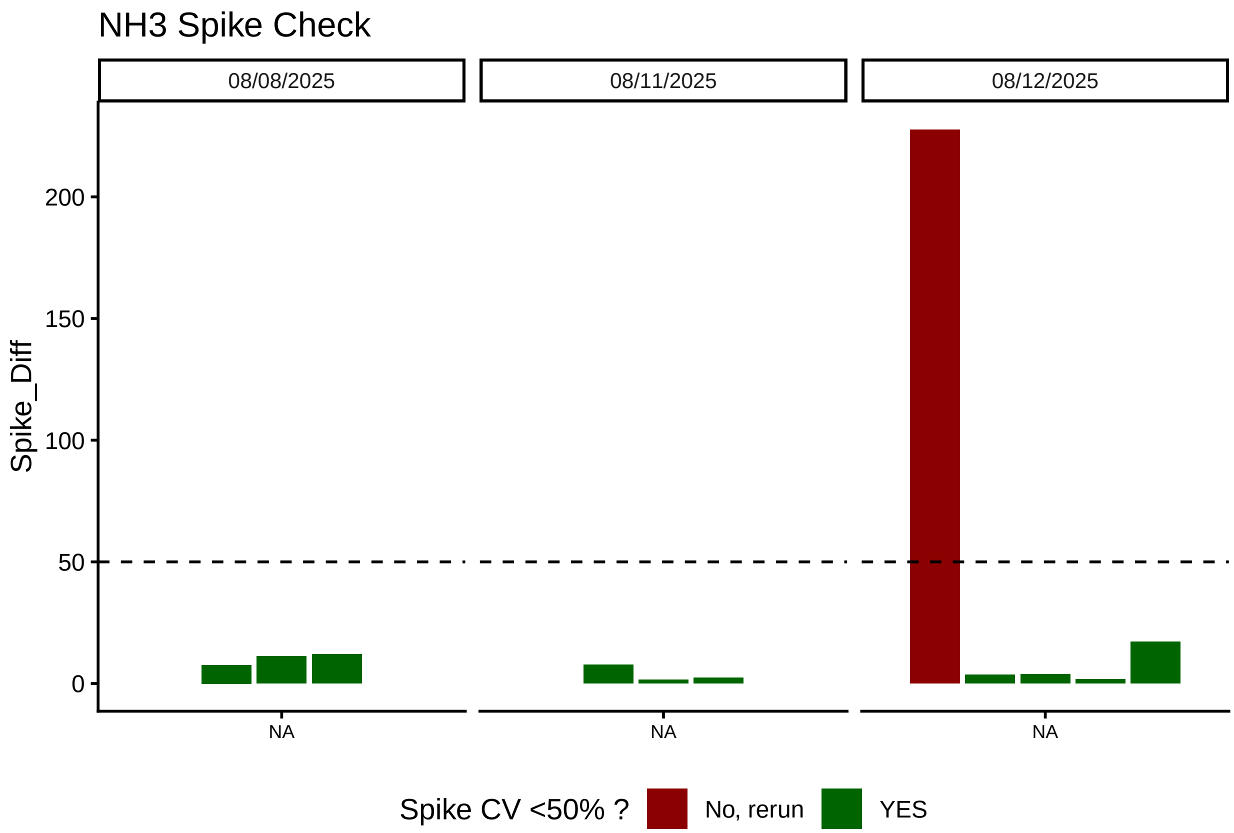


0.7 Analyze Duplicates



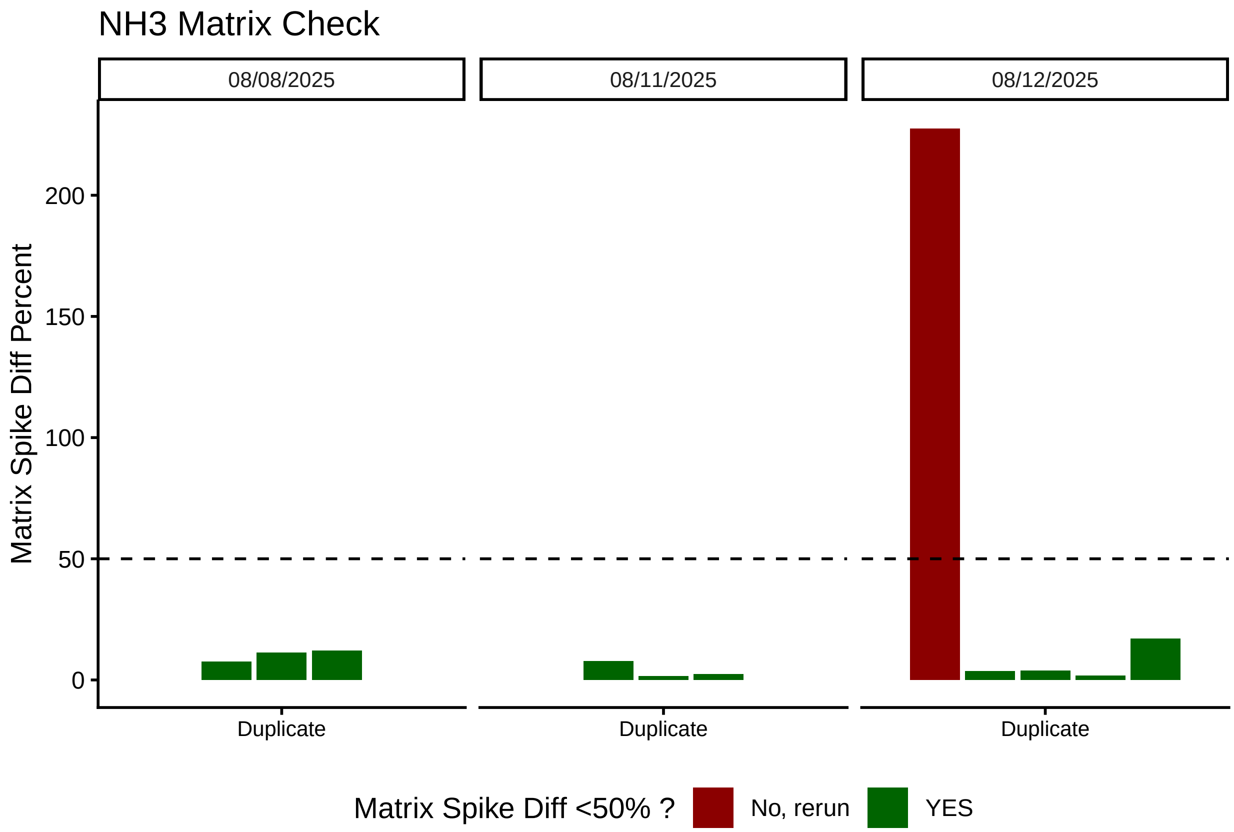
Test	Run_Date	Dup_Flags
Ammonia 2	08/08/2025	>60% of Dups Pass - PROCEED
Ammonia 2	08/11/2025	>60% of Dups Pass - PROCEED
Ammonia 2	08/12/2025	>60% of Dups Pass - PROCEED

0.8 Spikes



Test	Run_Date	Spike_Flags
Ammonia 2	08/08/2025	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	08/11/2025	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	08/12/2025	>60% of Spikes have a Diff <50% - PROCEED

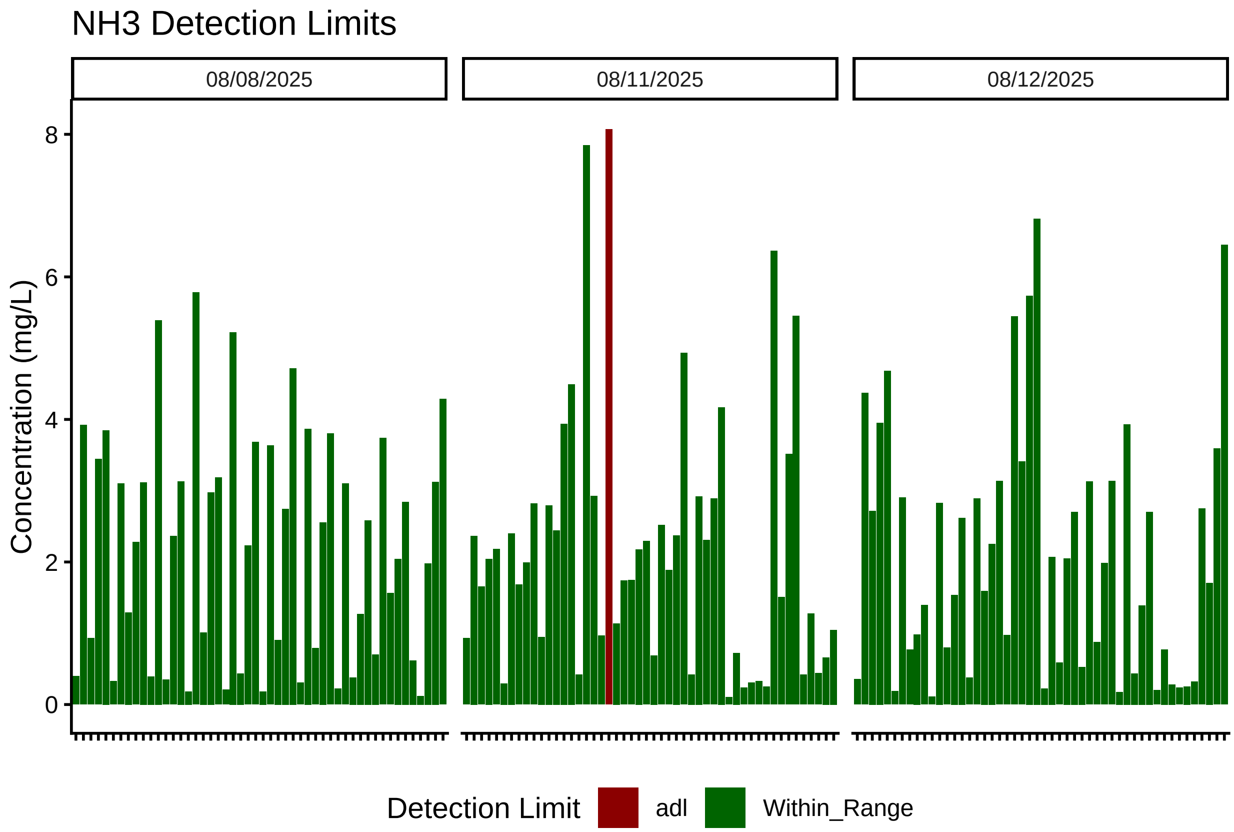
0.9 Matrix Effects



Test	Run_Date	Matrix_Flags
Ammonia 2	08/08/2025	Matrix Has Diff <50% - PROCEED
Ammonia 2	08/11/2025	Matrix Has Diff <50% - PROCEED
Ammonia 2	08/12/2025	Matrix Has Diff <50% - PROCEED

0.10 Unit Converted Data Column Added (mg/L to uM)

0.11 Sample Flagging - Within range of standard curve



##Add QAQC Flags

0.12 Create Log for QAQC

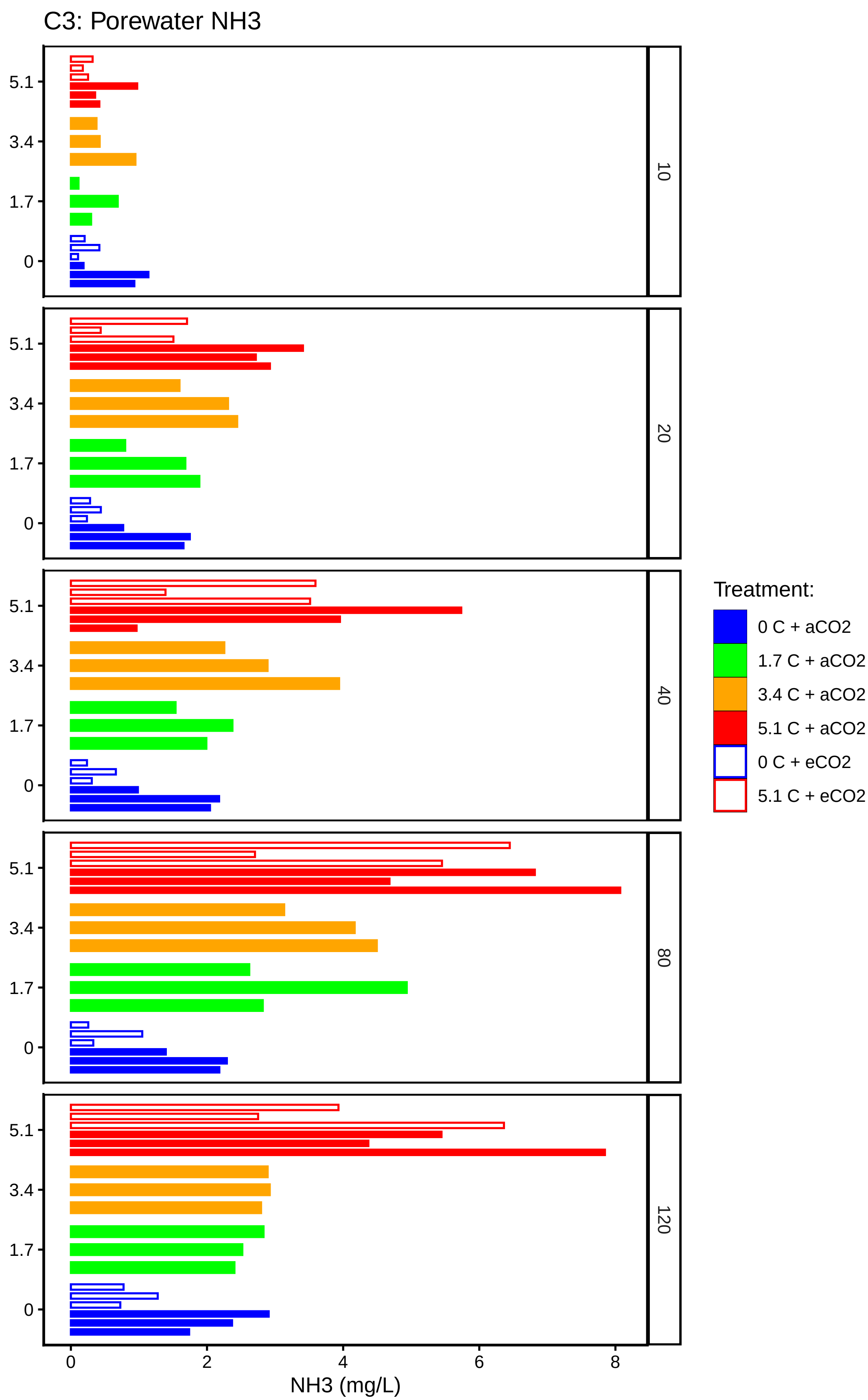
##Write QAQC Log

0.13 Merge samples with Metadata & check all samples are present

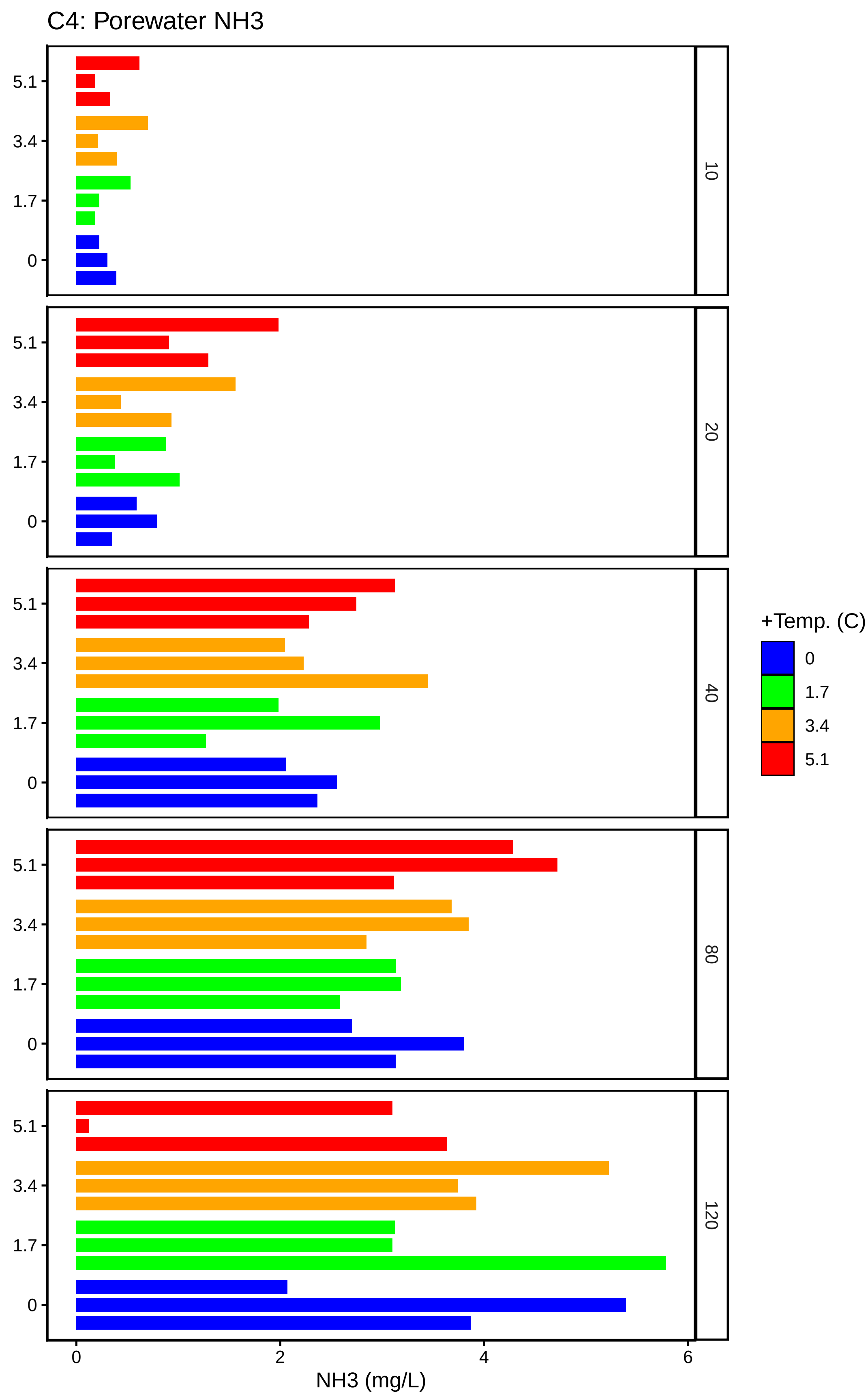
```
## Merge Samples with Metadata

## All sample IDs are present in data
```

0.14 Visualize Data

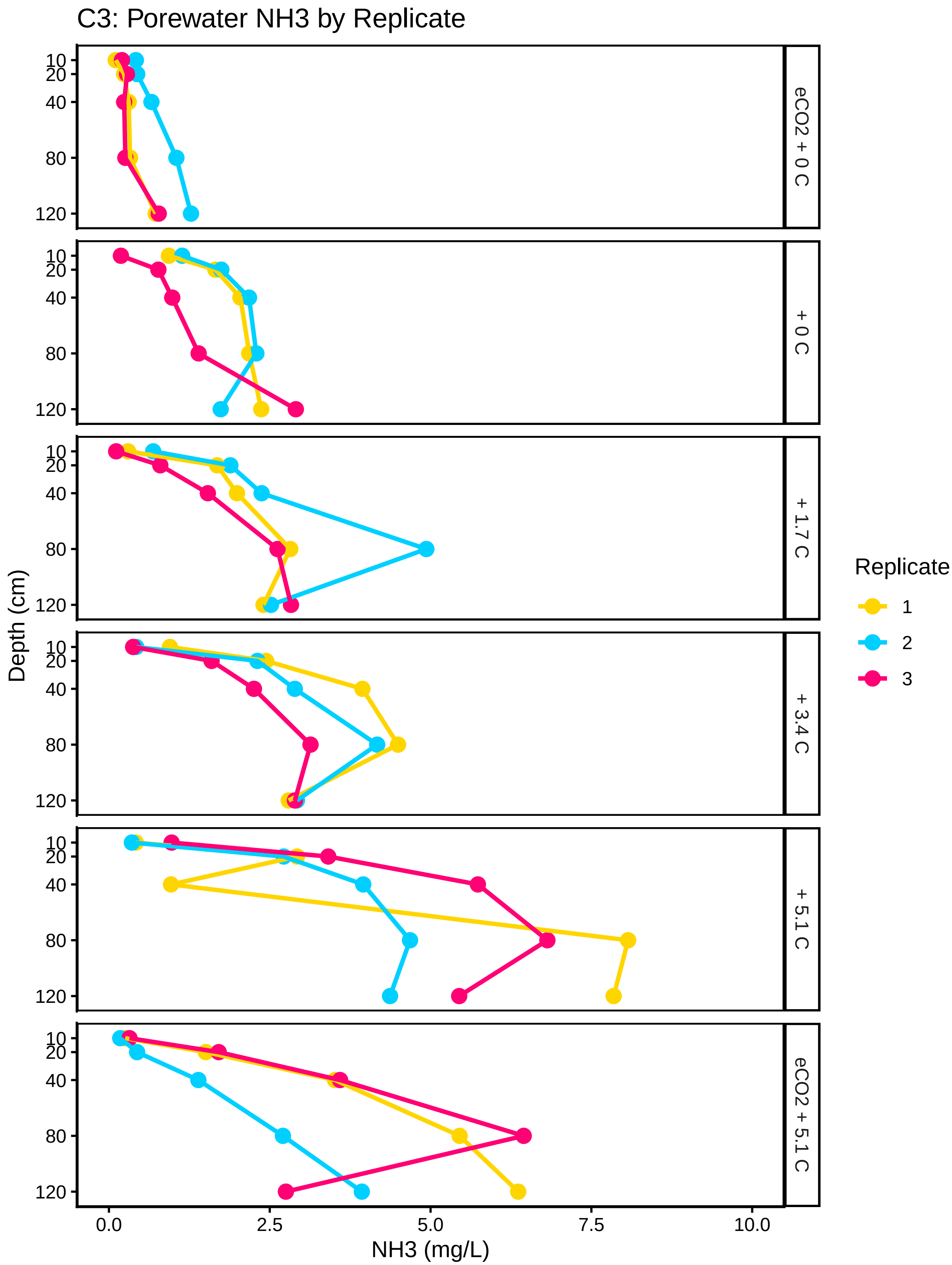
```
## Visualize Data
```



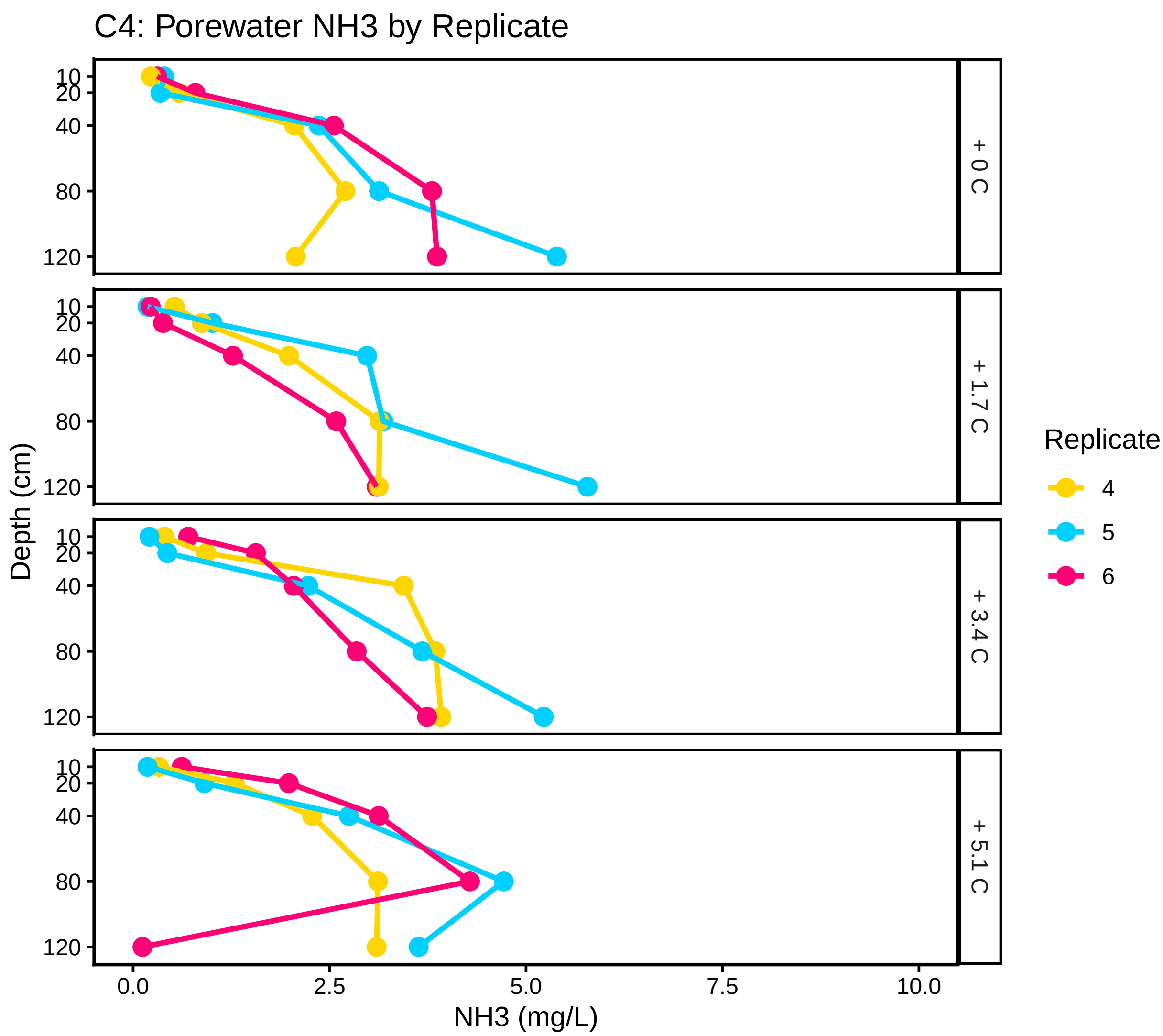
C3: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
2
4
6
8
NH3 (mg/L)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2


C4: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
2
4
6
NH3 (mg/L)
+Temp. (C)
0
1.7
3.4
5.1

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



C3: Porewater NH3 by Replicate
Depth (cm)
NH3 (mg/L)
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
Replicate
1
2
3
10
20
40
80
120
0.0
2.5
5.0
7.5
10.0



0.15 Export Processed Data

"'

#end